

Evaluating Ramp Meters: Evidence from the Twin Cities Ramp Meter Shut-off

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ABSTRACT

Ramp meters in the Twin Cities were turned off for 8 weeks in the Fall of 2000. This paper analyzes the results with and without ramp metering for two representative freeway, US 169 Northbound and I-94 eastbound during the afternoon peak period. Five performance measures: Accessibility, Mobility, Equity, Productivity and Consumer Surplus are compared. It is found that ramp meters are particularly helpful for long trips relative to short trips. On Route 169, that trips more than 3 exits in length benefit, while those shorter are hurt by ramp meters. Ramp metering, while generally beneficial to freeway segments, may not improve trip travel times (including ramp delays). The results are mixed, suggesting a more refined ramp control algorithm which explicitly considers ramp delay is in order.

Key Words: Ramp Meters, Evaluation, Equity, Mobility, Accessibility, Productivity, Consumer's Surplus

INTRODUCTION

In what may be the single most comprehensive experiment in the history of surface transportation, ramp meters in the Twin Cities have been the subject of a recent test of their effectiveness. By turning the meters off for 8 weeks in October, November, and December 2001, it has been possible to determine how well the metering system meets stated objectives, and to provide new insight into the appropriateness of those objectives. This report presents the detailed results of an analysis of observed data of two representative freeway, US 169 Northbound and I-94 eastbound during the afternoon peak period. and considers at a variety of measures of effectiveness resulting with and without ramp meters to control freeway traffic in the Twin Cities area. Earlier research (the papers Perspectives on Efficiency and Identifying Winners and Losers in Transportation submitted separately for presentation at this conference) identified a

number of alternative performance measures These included Accessibility, Mobility, Equity, Productivity and Consumer Surplus.

Ramp meters in the Twin Cities were intended "to optimize flow in metro area freeway corridors by making efficient use of available transportation facilities" (MnDOT 1996a). The Minnesota Department of Transportation made this goal operational through a control strategy that divided freeways into zones terminating at bottlenecks. The number of vehicles in each zone at a given time was constrained by the capacity of the bottleneck. Ramp metering was used to limit those vehicles.

However controversy arose when a State Senator from rural Minnesota challenged the conventional wisdom of the state Department of Transportation. The long delays at some ramps (at times, though not generally, in excess of 20 minutes) to ensure that the freeway remained free-flowing drew the ire of some commuters, who believed the system was at best inefficiently managed. The state legislature passed a bill in Spring 2000 requiring a ramp meter shut off experiment. This paper is an analysis of the data collected with and without metering.

The next section details the method used to measure travel times on ramps and freeway segments from the data available. The following section outlines the various performance measures and shows how they are computed. Then results are presented for Route 169 and I-94 for each of the performance measures. Recommendations and conclusions are delivered at the end of this report.

MEASURING TRAVEL TIMES

This section summarizes the calculation methodology required to measure travel times (and speeds) on entrance ramps, freeway segments, and O-D pairs on a highway with and without ramp meters.

The data collected for entrance ramps come in two types of paired data:

- Departure rate (Q_k)/ arrival rate (q_k) pair in each time interval (k) obtained from volume detectors.
- Departure rate (Q_k)/ queue length (n_k) pair for each time interval obtained from volume detectors and periodic visual (by remote camera) observation of queue length.

The second type data can be transformed to the first type by equation 1¹:

$$q_k = Q_k + n_k - n_{k-1} \quad (1)$$

Where:

- q_k the arrival rate in time interval k (vehicles/hour)
- Q_k the departure rate in time interval k (vehicles/hour)
- n_k the queue length in time interval k (number of vehicles)

Using the I/O queueing diagram shown in Figure 1, it is possible to find the delay for every individual vehicle: the delay of vehicle v is $(t_2 - t_1)$. To make the picture more clear, the method to calculate departure time for individual vehicles is given. The same method can be used to obtain arrival time, but assuming a uniform arrival at the back of the queue is not as accurate as assuming a uniform departure from the front of the metered queue, so will be discussed more below.

The data are collected in 5 minutes intervals (5 minutes = 300 seconds). So, the average time headway for time interval k equals $300/Q_k$ (sec). Then the departure time of the first vehicle is t_1 which is equal to h_1 .ⁱⁱ The departure time of the second vehicle is t_2 which is equal to $(h_1 + h_2)$. As the time headway is accumulated, the departure time (T_v) for every individual vehicle can be calculated. The arrival time (t_v) can also be obtained by the same methods. The delay (d_v) of every vehicle can be calculated by equation 2:

$$d_v = T_v - t_v \quad (2)$$

Where:

t_v arrival time for vehicle v
 T_v departure time of vehicle v
 d_v delay of vehicle v (sec)

The departure rate within the window is still assumed to be uniform due to the presence of ramp meters. An assumption must be made about the arrival rate of vehicles at the back of the queue. Observed data is only available in five minute traffic counts. A uniform arrival rate would give a lower bound for delay, a more reasonable assumption is to use an exponential (Poisson) arrival rate, which allows for bunching of vehicles. If the number of vehicles arriving at a queuing system has a Poisson distribution with a mean of "a" customers per unit of time, the time between arrivals has an exponential distribution with a mean of "1/a." With the data collected by departure rate detectors, arrival rate detectors and queue length cameras at these ramps, the ramp delay for each vehicle and the average delay in each 5-minute time intervals are obtained. The headway between vehicles is simulated using an exponential distribution 50 times, computing the delays using equation 2, and then averaging the delay.

Freeway loop detectors provide the volume and occupancy information in 5 minutes intervals, useful for computing traffic flow on freeway segments. Based on these data, the space mean speeds in every time interval can be computed as in equation 3:

$$\bar{u}_{skl} = \frac{100 * Q_{kl} * L_v}{K_{kl} * 300} \quad (3)$$

Where:

$\bar{u}_{skl}$ space mean speed of detector l in time interval k
 Q_{kl} volume of detector l in time interval k (vehicles/hour)
 L_v average vehicle length plus the length of the loop detector
 K_{kl} time occupancy of the 1st (2nd) detector in time interval k (%)

Because Route 169 is a two-lane freeway, there are two loop detectors at each station, there are two space mean speeds. The weighted mean of the two lanes will be used as the speed at each station, which is:

$$\bar{u}_{sk} = \frac{Q_{k1} * \bar{u}_{sk1} + Q_{k2} * \bar{u}_{sk2}}{Q_{k1} + Q_{k2}} \quad (4)$$

Assume that this speed is the average speed within this segment (that is, speeds within a segment are uniform). If there are no detectors in a segment, the speed of the nearest station will be assumed as the average speed in this segment. Then the travel time of every time interval can be obtained:

$$\tau_k = L / \bar{u}_{sk} \quad (5)$$

Where:

τ_k travel time in time interval k

L Length of section

The free flow travel time (F) of a specified freeway segment is defined as the shortest travel time on this segment of all the five minute intervals (k).

$$F = \min(\tau_k) \quad (6)$$

Delay during time interval k is defined as the difference between free flow travel time (F) and actual travel time (τ_k).ⁱⁱⁱ

$$d_k = \tau_k - F \quad (7)$$

Once the average delay on ramps and the average travel time in freeway segments in each time intervals are obtained, it is possible to build the travel time O-D matrices. However, this requires that the data be synchronized. First a database which records the delay and travel time data for each five minute period is built. Then the travel time for each O-D pair (from ramp i to ramp j) will be calculated. Similarly, a delay matrix can be constructed. The travel time is calculated with equation 8:

$$\tau_{i,j,k} = d_{r,k} + \tau_{i,k+x1} + \tau_{i+1,k+x2} + \dots + \tau_{j-1,k+x(j-i)} \quad (8)$$

Where:

$\tau_{i,j,k}$ travel time from origin i to destination j departing in time interval k;

$d_{r,k}$ delay on ramp r in time interval k;

τ_{k+x1} travel time in the freeway segment i in time interval (k+x₁);

$x_1, x_2, \dots, x_{j-1}$: synchronization coefficient, x is equal to the integer part of the result of the travel time from the origin to the beginning of this segment divided by the time interval (300sec).

PERFORMANCE MEASURES

A number of performance measures are used to evaluate the ramp meters. Mobility, or ease of movement can be measured on ramps, freeway segments and for the system as a whole. Total delay, number of vehicles being delayed, average delay through the whole observation period, and average delay of each time interval are computed for each ramp. The travel time and delay for each freeway segment are measured. They are combined through synchronization into a series of OD matrices containing different mobility measures (travel time, travel delay, speed).

The Gini Coefficient of concentration and the Lorenz curve have been used in economics to analyze income inequality, they are applied here to analyze travel time, delay, and speed inequalities. The distribution of the total delay is represented by the Lorenz curve and the statistical analysis is done using the Gini coefficient, a measure of inequality. Figure 2 illustrates the Lorenz Curve for the case of ramp 3A1, the line separating A_1 and A_2 (the striped (green) and light (yellow) areas respectively). The Lorenz curve relates the proportion of the population receiving a given proportion of delay. While the bottom 100% of the population gets 100% of delay by definition, the bottom 50% may only get 30% of the delay. A Gini Coefficient of 0.0 indicates perfect equality, and 1.0 indicates perfect inequality (the most delayed person suffers all the delay). The Gini Coefficient (G) is defined as

$$G = A_1 / (A_1 + A_2) \quad (9)$$

Where:

A_1 area between Lorenz Curve and 45-degree line

A_2 rest of triangle defined by Lorenz Curve, x-axis, and vertical line projecting from 100% of the population in question.

For each mobility measure OD matrix, corresponding spatial equity, temporal equity and temporal-spatial equity are evaluated. The equity analysis is illustrated for the speed OD matrix, shown in Table 2. If the equity measures are calculated by the speeds in all the time intervals for just one OD pair (e.g. speed 11, speed 21, ... , speed m_1), the temporal equity can be evaluated. These temporal equity measurements measure the differences between drivers who travel on the same route but arrive at the origin (entrance ramp) at different times. On the other hand, if the equity measures are obtained with respect to the speed in the same time intervals but different OD pairs (e.g. speed 11, speed 12, ... , speed $1n$), spatial equity results. These equity measures tell the differences between drivers who arrive at the different entrance ramps at the same time. Also, to evaluate the equity for the whole network throughout the observation period, the temporal-spatial equity measures can be used which are calculated by all the $m \times n$ speed values.

Consumer surplus measures the difference between the market price of a particular good and the price the buyer is willing to pay for that good. Because the willingness to pay varies between individuals, the consumer surplus for two persons will not be the same. However, consumer surplus can be aggregated across individuals. In general, change in consumer surplus is used rather than as absolute measure, because it is more readily measured. In travel behavior, the price in this case is the travel time while the quantity of the good is the number of vehicles using the road network. As the travel time across a certain stretch of the traffic network decreases there will be more vehicles willing to access the same stretch pushing the flow up. Mathematically, this can be approximated as:

$$\Delta CS = 0.5(Q_{off} + Q_{on})(\tau_{off} - \tau_{on}) \quad (10)$$

Where:

Q_{on}, Q_{off} flows when the ramp meters are on, off respectively.
 τ_{on}, τ_{off} travel times when the ramp meters are on, off respectively.

If the travel time when the ramp meters are switched off is greater than when they are switched on, then there is a loss of consumer surplus with metering.

Productivity is the ratio of the output of any product to the input that was required to produce that output. For transportation networks, vehicle kilometers traveled (VKT) and the vehicle hours traveled (VHT) are the output and the input respectively.

$$P = \frac{\text{output}}{\text{input}} = \frac{\sum VKT}{\sum VHT} = \frac{\sum Q * L}{\sum Q * \tau} \quad (11)$$

The ratio of VKT to VHT is measured for each freeway segment and ramps separately and then added to obtain the productivity of the system for both the ramp meter on and ramp meter off cases. The higher the productivity with ramp metering, the better the ramp metering is.

From the transportation planner's perspective, accessibility is defined as the ease of reaching destinations. Weibull (1980) suggests that accessibility is a measure of an individual's freedom to participate in activities in the environment. Areas with high congestion often have high accessibility. Three different functions of travel time are used, these are described in the results section. Freeway accessibility is computed both with and without ramp metering.

$$A_i = \sum_j Q_j \times f(\tau_{ij}) \quad (12)$$

where:

A_i accessibility of zone i
 Q_j opportunities at zone j, measured using exit volumes

τ_{ij} travel time between i and j

RESULTS

Route 169 Northbound is two lanes in each direction, divided into 16 freeway segments that are served by 16 ramps. The analysis section begins at Valley View Road and continues until Medicine Lake Road. However for the segment between TH 62 EB and TH 62 WB, the data was unavailable and so the ramps were reduced to fifteen in total. The data for the analysis include the 5 minute flows and travel times on the freeway segments and the ramps for both the metered and the unmetered cases. The date, times (2:55 p.m. - 6:10 p.m.), and locations of this analysis are the same across all performance measures for Route 169. In order to make the results comparable, the data used for the analyses (ramp metering on and off) were collected on Tuesdays: March 21 and November 7, 2000. November 7 is the third Tuesday after ramp lights were shut down. This assumes that an equilibrium was established and that traffic patterns were stable after three weeks. Entrance ramp delays are only analyzed for March 21, as in the absence of ramp metering there should be no significant ramp delay. However ramp volumes for both dates are considered.

First the relationship between mobility and equity for O-D pairs on Route 169 with and without ramp meters is estimated. Figures 3 and 4 illustrate the trends in the change of the mobility and equity with and without metering. Note that in Figure 3, the shortest trips (those on the right side of the graph) actually are hurt in mobility terms by ramp metering, while the longest trips, (those on the left side) benefit the most. Figure 4 looks at the equity across time periods, and shows improving equity and worsening mobility as the peak is reached. Both results suggest that equity is increased when ramp metering is off.

What ramp meters delivers in term of mobility and equity can be shown by the comparison of the two cases in Table 2. Previous research indicates that ramp meters can increase the mobility of freeway networks, which is confirmed here. With ramp metering, the average travel speed (taking ramp delay into account) of the network increases from 37 to 62 km/hour; travel delay per km decreases from 82 sec to 68 sec and the average travel time for one trip decreased from 610 sec to 330 sec. No previous results can be relied to guide the analysis of equity. When looking at trips a drop in the Gini coefficient in the absence of metering is found. This suggests the system becomes more fair when meters are removed. This drop is observed for three primary measures: travel time per mile, travel speed, and travel delay per km, as shown in Table 2.

Table 3 shows productivity, the vehicle kilometers of travel, vehicle seconds of travel on freeway segments and ramps. The freeway segments have productivity of 102 km/hour with metering and 52 km/hour without meters. The changes in consumer surplus for each individual segment are summed to get a benefit from metering of 3531 vehicle hours. The net productivity of the ramps themselves is 5.76 km/hour with meters and (by assumption) 40 km/hour without meters. The change in consumer surplus is

found to be -639 vehicle hours. As expected, ramp meters significantly reduce the productivity and consumer surplus of the ramps.

Combining freeway segments with ramps gives a system productivity measure. The system productivity improves immensely with ramp metering. In fact the percentage increase in system productivity is 64%. The change in consumer surplus of the system with ramp metering is overall positive, the ramp meters benefit the freeway segments more than they harm the ramp segments, so an overall positive change in consumer's surplus of 2893 vehicle hours is recorded.

Three different accessibility models were applied to Route 169. The first is a classic gravity model, the second a model estimated for freeways in the Twin Cities, and the third from a regional gravity model estimated by the author for Washington DC (Levinson and Kumar 1995). For all the three cases, accessibility increases with the ramp metering, as shown in Table 4. However, there may be accessibility functions for which this is not the case.

I-94 east bound is divided into eight freeway segments from Olson Highway (TH55) to Marion St. which consists of nine entrance ramps and eight freeway segments. For the most part, the I-94 is 3-lanes except from Cretin avenue to Dale street where it is 4-lanes. Again, we are working with the afternoon peak period from 2:55 p.m. - 6:10 p.m. Analyzing I-94 paints a somewhat different picture than Route 169. In order to make the results more comparable, the data used to calculate in both cases were collected on Wednesdays (03/29/00 and 11/01/00 respectively). Since November 1 is the third Wednesday after ramp lights were shut down, it is assumed that a new traffic equilibrium was established and the whole network was stable.

Surprisingly, on I-94, the network mobility measures decrease slightly as the result of the ramp meter control. The average travel speed (taking ramp delay into account) of the network decreases from 87 km/hour (without control) to 79 km/hour (with control). Travel delay per mile increase from 27.9 sec (without control) to 42.1 sec (with control) and the average travel time for one trip increase from 285 sec (without control) to 299sec (with control). In terms of temporal-spatial (network) equity: the ramp metering off case is more equitable. Because the ramp off case has higher mobility, there is a negative relationship between mobility and equity.

Figure 5 shows temporal equity: overall ramp metering is less efficient and equitable than no metering. Also, the different effect of ramp control on long trips and short trips can be demonstrated by the figure $\frac{3}{4}$ long trips are better off, short trips are worse off with metering. Figure 6 illustrates spatial equity: it seems that the relationship between spatial equity and mobility is positive, but this is not always the case.

Despite the seeming poor performance on the mobility measure, weighting the data differently, in Table 3, suggests an increase in the productivity of the system by 8.26%. The net change in the consumer surplus of the whole system including freeways (683 vehicle hours) and ramps (-201 vehicle hours) is 481 vehicle hours.

The accessibility measures for I-94 are also shown in Table 4. Unlike Route 169, but consistent with our mobility measures, these results are mixed. In one of the three cases, accessibility falls with metering.

On I-94, there is an issue regarding the location of the study section. In the study section considered, metering rates were strict to minimize the formation of bottlenecks farther downstream. If these downstream locations and the ramps leading to them were accounted for in the calculation process, then there might have been variation in the results. However that data (particularly the ramp queues with metering) was unavailable.

CONCLUSIONS

This paper dealt with a number of measures of effectiveness. In general, the findings were favorable to ramp metering on Route 169. These findings are limited in scope as they are based on only one representative day of each of the metering and no-metering period. Different sections change the results. Analysis of I-94 for instance, tempers some of the glowing results for ramp metering, suggesting a larger proportion of O-D pairs are worse off with meters than without, though other overall measures of effectiveness are still generally positive.

Prior to October 2000, the Minnesota ramp meter control strategy had been focused on freeway segments, ignoring delay on ramps. The theory of freeway ramp metering argues that ensuring higher flows on freeways guarantees lower flows elsewhere on the road network. The more vehicles able to use the freeways, the fewer that will burden local streets. By the objectives the ramp meters were intended to satisfy, maximizing freeway throughput at selected bottlenecks, this study of Highway 169 between I-494 and I-94 confirms that the facility performs better in the presence of operating ramp meters than in their absence. The change in consumer surplus was positive while at the same time, productivity of the system almost doubled. Freeway speeds and flows are consistently higher with ramp metering on than without. Furthermore, Route 169 shows better trip speeds with metering. While this is logical if the intention is to maximize freeway throughput, it does not necessarily maximize user satisfaction of the system. In contrast, I-94 shows improvements to the operations of freeway sections do not always offset the additional ramp delay. Metering must be used finely with a chisel, rather than coarsely with a sledge-hammer.

Ramp meters are particularly helpful for long trips relative to short trips. This study of Route 169 shows that trips more than 3 exits in length benefit, while those 3 exits or less are hurt by ramp meters. Originally freeways (the *interstate* highway system) were designed to serve intercity not local traffic. However freeways have evolved to serve commuters. Whether reserving the freeways for long trips is still an appropriate objective is an important public policy debate.

Other possible objectives or constraints, such as ensuring that delay is distributed as evenly as possible, are not achieved by the ramp metering control in place before this

study. Consider for instance traffic signals on streets. If a minor road meets a major road, and the major road is operating at capacity, it might be most efficient (in terms of minimal total delay) to give 100% of the green time to the major road and 0% to the minor road. This is not done. Traffic signals alternate back and forth in part to ensure some equity, so that travelers on the minor road do not have an excessive wait. Generally cycle lengths on roads at capacity are under 3 minutes, though this varies. A similar limit on individual delay, even at the expense of overall freeway efficiency, may be necessary for ramp meters to satisfy equity considerations. Future research will pursue alternative strategies. Furthermore, if efficiency is defined more broadly, for instance, maximizing the utility of travelers, recognizing a non-linear value of time, that constraint may be even more essential. So while the Minnesota algorithm may in generally be better than no metering at all, there may still be other control strategies that perform better still on a wider variety of performance measures.

We believe that a ramp metering system that satisfies users must consider ramp delay in addition to freeway throughput. Unfortunately current data does not permit any such strategies. Aside from a few spotty locations, there exists no accurate measure of the number of vehicles waiting in queue at a ramp meter at any given time in the Twin Cities. The study attached here uses data from special data collection efforts carried out by Mn/DOT's Traffic Management Center on a few freeway facilities. Unfortunately, such data collection efforts that are not routine. Additional data collection systems to measure queue lengths in real time are required. Also Origin-Destination data would enable better algorithms as well as better measures of effectiveness.

Finally, it must be remembered that ramp meters can not be expected to carry the entire load of traffic growth. Given the under-investment in highway capacity relative to growth in demand, and the present unwillingness to effect demand through pricing or other measures, congestion is inevitable. While ramp meters can help at the margins, delaying the onset of freeway breakdowns and by making freeways flow smoother, they can not eliminate congestion entirely.

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TABLES AND FIGURES

Table 1 Speed OD Matrix

Speed	OD pair 1	OD pair 2	...	OD pair n
Time Interval 1	Speed 11	Speed 12	...	Speed 1n
Time Interval 2	Speed 21	Speed 22	...	Speed 2n
...	...	...	...	...
Time Interval m	Speed m1	Speed m2	...	Speed mn

Table 2: Temporal-Spatial Equity and Mobility

			Travel Speed	Travel Time per km	Travel Delay per km
With Meters	Route 169	Mobility Measures	62 km/hour	99 sec	68 sec
		Gini Coefficient	0.28	0.46	0.68
		<i>I-94</i>	<i>79 km/hour</i>	<i>89 sec</i>	<i>42 sec</i>
	<i>I-94</i>	<i>Mobility Measures</i>	<i>0.17</i>	<i>0.24</i>	<i>0.49</i>
		<i>Gini Coefficient</i>			
Without Meters	Route 169	Mobility Measures	37 km/hour	113 sec	82 sec
		Gini Coefficient	0.23	0.21	0.28
		<i>I-94</i>	<i>87 km/hour</i>	<i>75 sec</i>	<i>28 sec</i>
	<i>I-94</i>	<i>Mobility Measures</i>	<i>0.14</i>	<i>0.16</i>	<i>0.39</i>
		<i>Gini Coefficient</i>			

Table 3: Productivity

		Segment		VKT/ VHT (km/hour)	Ramps VKT	VHT	VKT/ VHT (km/hour)	System Productivity (km/hour)
With meters	Route 169	339822	3341	102	3994	703	5.7	85
	I-94	539286	5494	98	3785	264	14.3	94
Without meters	Route 169	271388	5214	52	3815	95	40*	52
	I-94	523027	5940	88	3819	95	40*	87

*Note: VKT: Vehicle Kilometers of Travel, VHT: Vehicle Hours of Travel, * by assumption*

Table 4. Accessibility measure

Model	With Meters	Without Meters	% Change	<i>With Meters</i>	<i>Without Meters</i>	% <i>Change</i>
	Route 169			<i>I-94</i>		
$f(C_{ij}) = 1/t_{ij}^2$	2.9	2.5	16%	12.5	8.5	46 %
$f(C_{ij}) = e^{-0.00189 \cdot t_{ij}}$	58932	34502	71 %	42977	43597	-1.4 %
$f(C_{ij}) = e^{-0.08 \cdot t_{ij}}$	93.2	63.772	46%	822.5	456.9	80 %

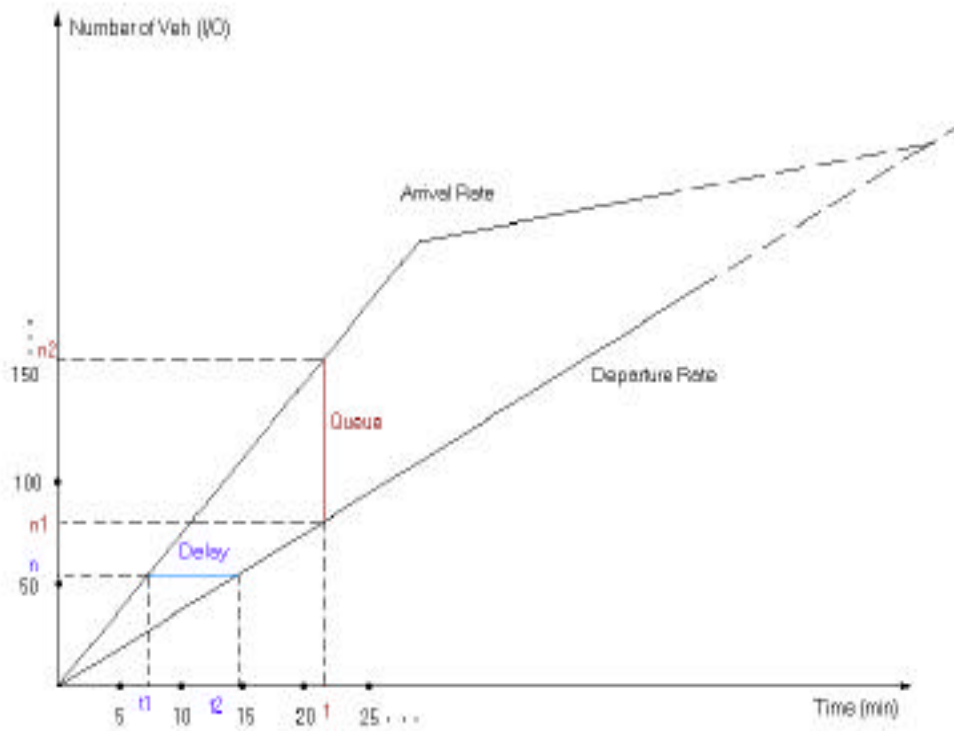


Figure 1: Queueing Diagram

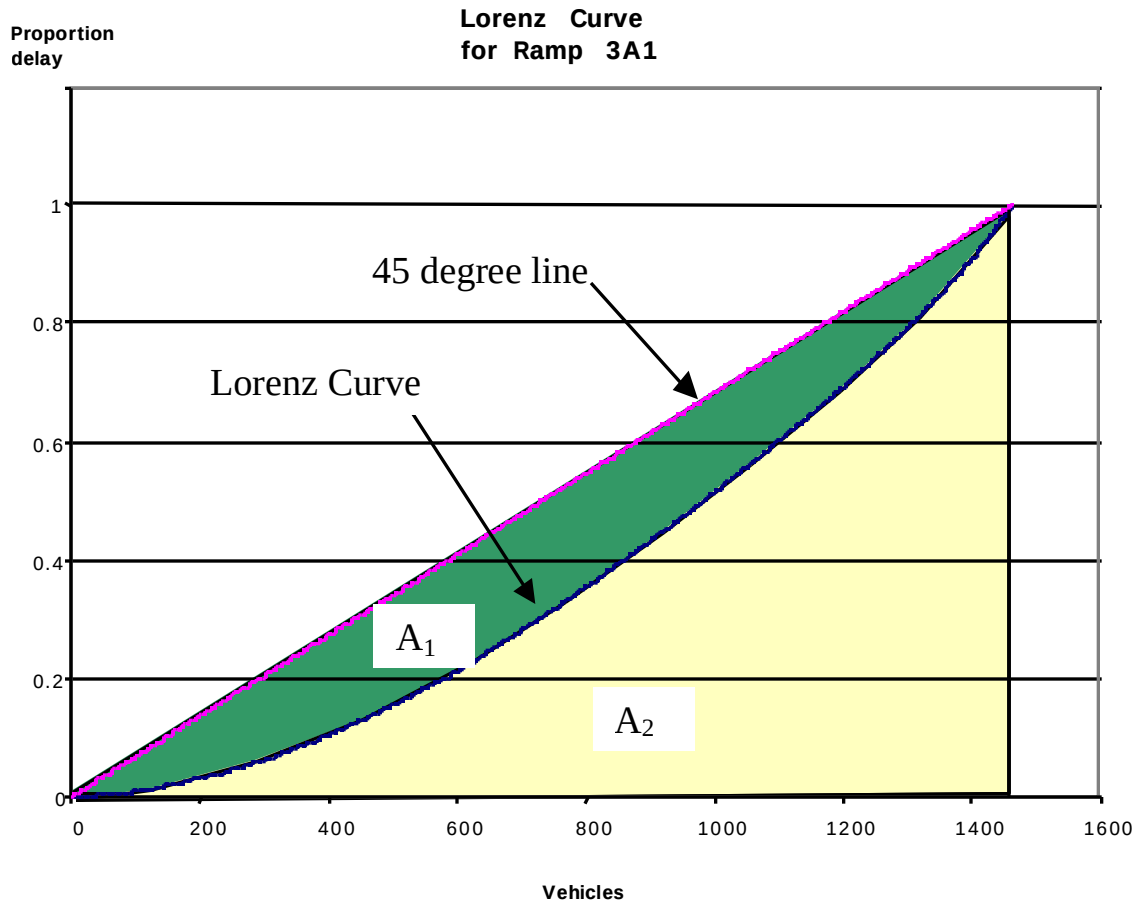


Figure 2: Lorenz Curve for Ramp 3A1, Route 169

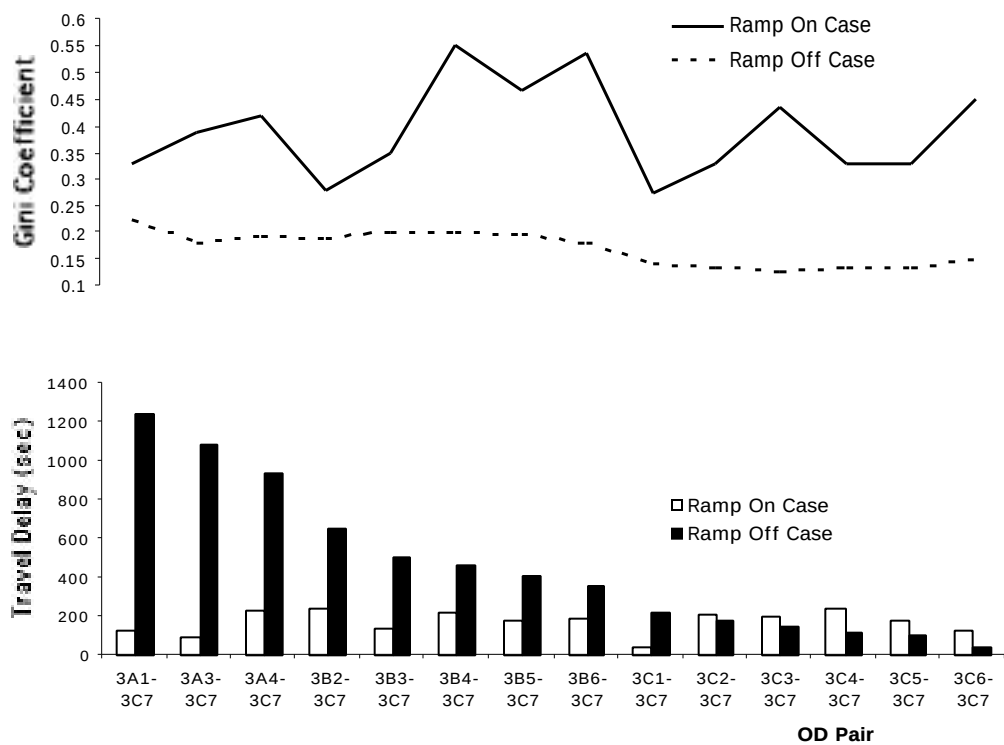


Figure 3: Temporal Equity and Mobility (Travel Delay), Route 169

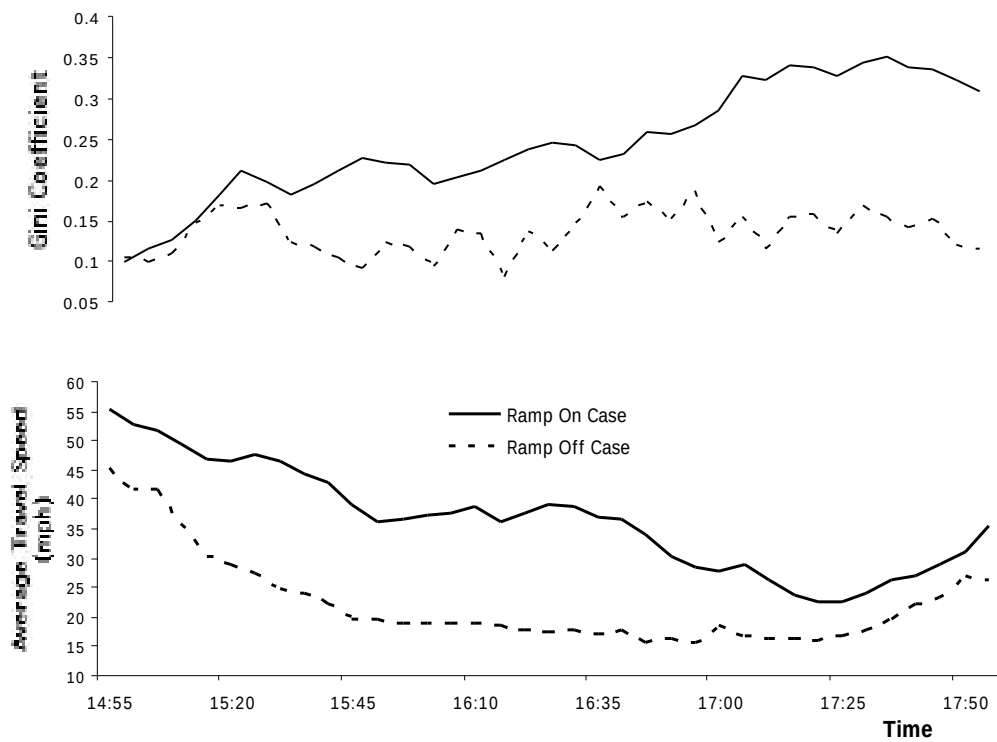


Figure 4: Spatial Equity and Mobility (Speed) Route 169

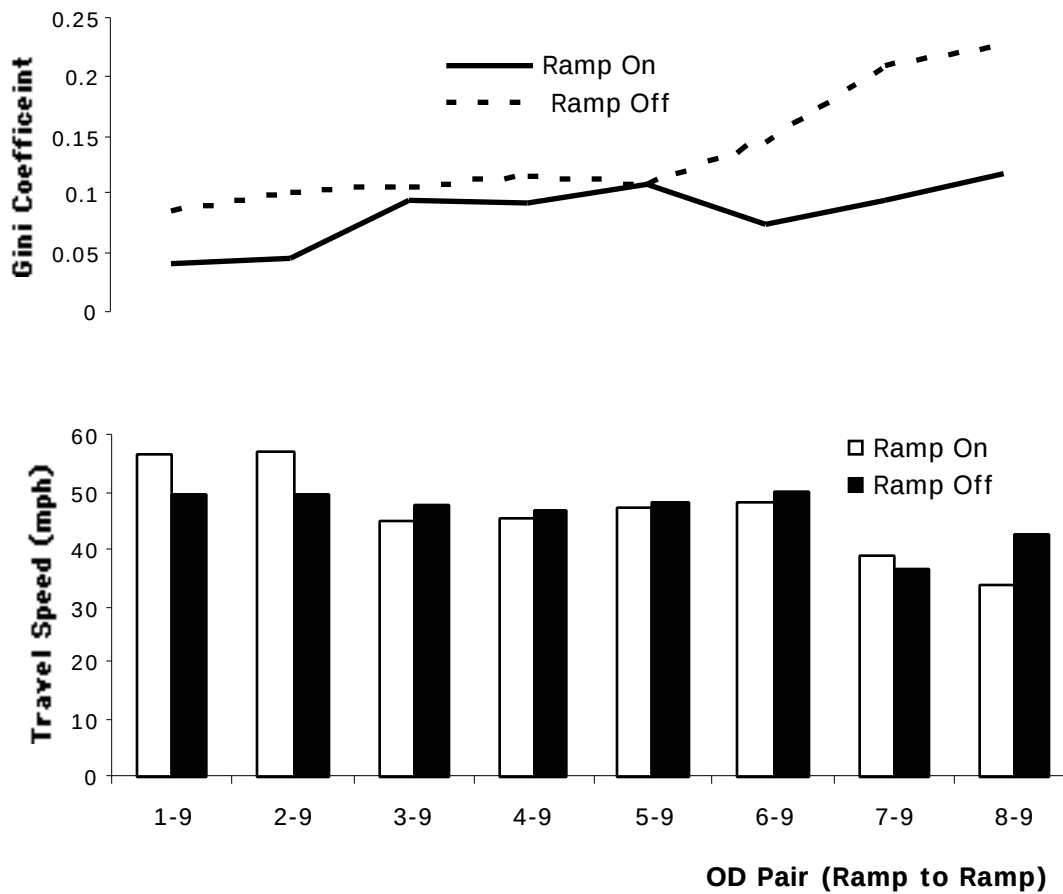


Figure 5: Temporal Equity and Mobility (Speed) I-94

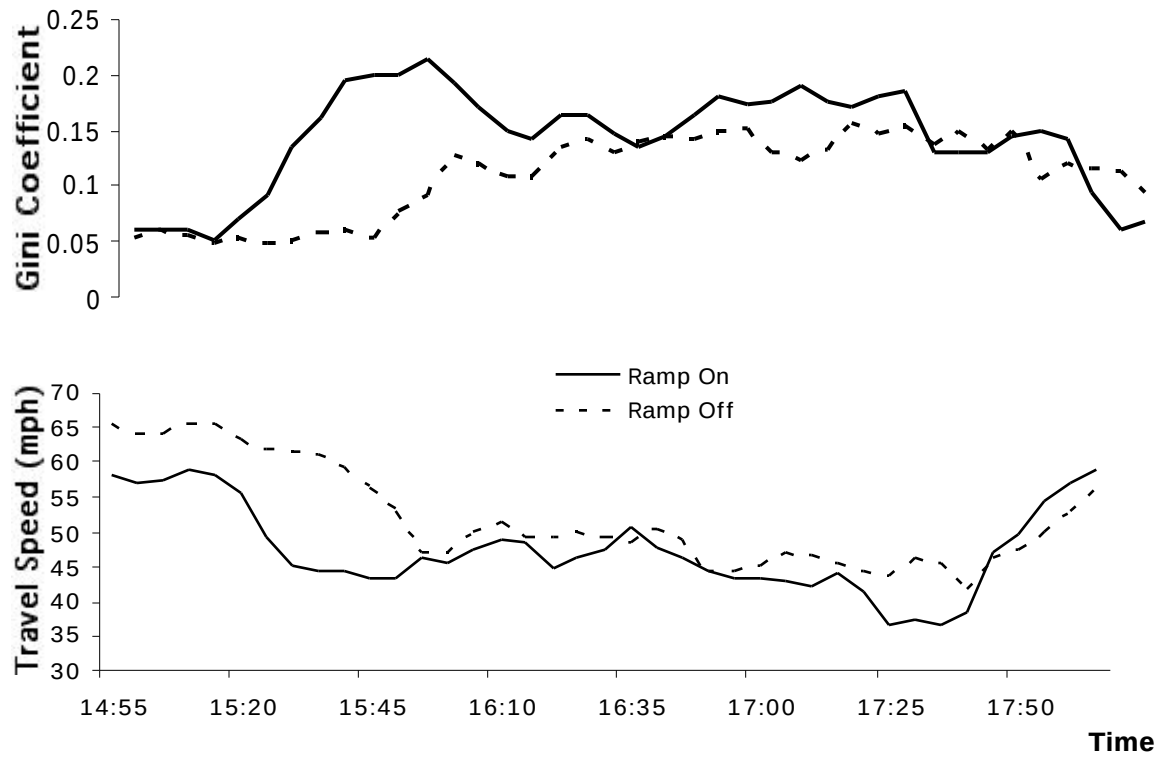


Figure 6: Spatial Equity and Mobility (Speed) I-94

ⁱ This transformation will make the results differ slightly from the results calculated by initial departure rate/ arrival rate pair data. If the departure rate/ queue length pair data is used to calculate delay, the delay will be the stopped delay under the ramp lights. However, if the departure rate/ arrival rate pair data is used, the delay will be the sum of the stopped delay and the travel time from arrival rate detector to the departure rate detector. Hopefully this discrepancy is minor as ramps are typically short. If the delay were calculated using only one type data, the results would be more comparable. Unfortunately, on some ramps, there are no arrival rate detectors or the arrival rate detectors don't give correct data, for instance, they double count some vehicles or miss others.

ⁱⁱ Alternatively, the departure time of a vehicle can also be the beginning of the corresponding headway. The difference between these two methods is small because the difference will not be accumulated.

ⁱⁱⁱ Alternatively freeway delay could have been defined as the difference between the actual travel time and the desired travel time (the travel time with the speed limit). In that case, reducing speed from 120 km/hour to 110 km/hour increases travel time, but still counts as zero delay, as the speed remains above the speed limit.